TaskFlow — Intelligent Task Management System

Final Project Academic Report | Introduction to Functional Programming

IT-2510

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https://github.com/azamatbukharbayev/IFPFinal

Executive Summary

TaskFlow is an enterprise-grade, responsive task and project management web application developed in C# 13 and .NET 10. The primary pedagogical and technical goal of TaskFlow is to demonstrate how modern functional programming paradigms—such as algebraic data types, monadic error handling (`Option<T>` and `Result<T>`), currying, partial application, function composition, graph traversal, and stack-safe trampolined recursion—can be elegantly married with enterprise Clean Architecture (Onion Architecture) and interactive Blazor Server user interfaces.

TaskFlow eliminates common software vulnerabilities such as null reference exceptions, uncaught runtime exceptions, and mutable state concurrency hazards through an isolated, referentially transparent Functional Core surrounded by an Imperative Shell.

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1. Introduction and Problem Statement

Contemporary students, researchers, and software engineering teams face escalating cognitive loads when attempting to coordinate overlapping deadlines, prioritize inter-dependent responsibilities, and monitor iterative progress across multiple concurrent initiatives. Commercial productivity applications (such as Jira, Trello, or Asana) often present rigid workflows or suffer from arbitrary task prioritization mechanisms that do not mathematically balance urgency against strategic importance.

TaskFlow addresses these challenges through an open, mathematically grounded task orchestration environment. The system features:

- Centralized Work Management: Full CRUD operations across tasks, subtasks, and long-term project containers.
- Dynamic Task Dependency Resolution: Automatic prevention of circular deadlocks and strict enforcement of prerequisite completion before dependent tasks can be finalized.
- Calculated Urgency Engine: Algorithmic calculation of dynamic priority scores rather than subjective, static priority tags.
- Multi-Perspective Visualizations: Dynamic KPI dashboard, 4-column reactive Kanban board, monthly deadline calendar, and detailed velocity analytics.

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2. Academic Objectives and Syllabus Alignment

The implementation strictly satisfies all topics outlined in the Introduction to Functional Programming university curriculum:

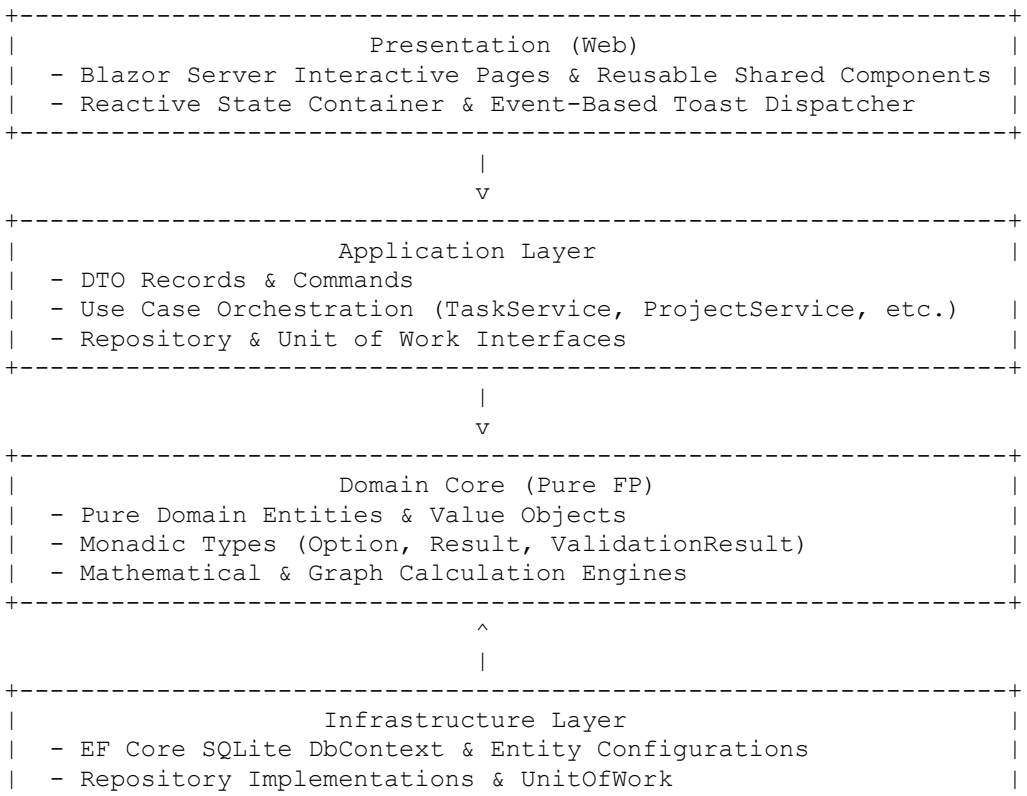
```
+-------------------------------------------------------------------------
---------+
| Syllabus Topic                   | Implementation Artifact
|
+-------------------------------------------------------------------------
---------+
| Pure Functions                   | TaskPrioritizationService,
```

```
DependencyGraph     |
| Immutability & Value Objects     | PriorityScore, Effort, Label,
DateRange        |
| Option / Maybe Monad             | TaskFlow.Domain.Functional.Option<T>
|
| Result / Either Monad            | TaskFlow.Domain.Functional.Result<T>
|
| Currying & Partial Application   | FunctionalExtensions.Curry, Partial
|
| Function Composition (Pipelines) | FunctionalExtensions.Then,
TaskAnalyticsService|
| Pattern Matching                 | Discriminated Unions
(TaskOperationResult)     |
| Recursion & Trampolining         | ProjectProgressService (Bounce, Done,
More)    |
| Memoization                      | FunctionalExtensions.Memoize
|
| Monadic List Processing (LINQ)   | TaskCollectionExtensions
|
| Directed Graph Traversal (DAG)   | DFS Cycle Detection & Kahn's Algorithm
|
+-------------------------------------------------------------------------
---------+
```

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3. System Architecture and Design Patterns

TaskFlow follows Clean Architecture principles, enforcing the Dependency Inversion Principle (DIP) where dependencies point strictly inward toward the core domain.



```
|  - Custom RFC-4180 CSV Engine & JSON Import/Export              |
+-----------------------------------------------------------------+
```

The Functional Core / Imperative Shell Separation

- Functional Core (`TaskFlow.Domain`): Purely deterministic algorithms, immutable state, and monadic transformations. Free from any I/O, database, or network libraries.
- Imperative Shell (`TaskFlow.Application` & `TaskFlow.Infrastructure`): Responsible for fetching data from persistent storage (SQLite), dispatching domain events to activity logs, handling time boundaries (`DateTime.UtcNow`), and file streams.

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4. Functional Core: Mathematical Formulation and Implementation

4.1. Monadic Constructs: `Option<T>` and `Result<T>`

To avoid the notorious billion-dollar mistake (null references) and prevent unstructured runtime exception crashes, TaskFlow implements complete, self-contained monads adhering to category-theoretic laws:

The `Option<T>` Monad

Models the presence or absence of a value without nulls:

- `Some<T>(T Value)`: Represents presence.
- `None<T>()`: Represents absence.
- Operators: `Map<TResult>`, `Bind<TResult>`, `Where`, `Match<TResult>`.

The `Result<T>` Monad & Railway-Oriented Programming

Models success or failure, accumulating error strings:

- `Success<T>(T Value)`: Successful execution.
- `Failure<T>(IReadOnlyList<string> Errors)`: Failed operation.
- Monadic combinator: `Combine<T1, T2>(first, second)` allows applicative validation, accumulating all validation errors rather than failing fast on the first error.

4.2. Mathematical Multi-Factor Prioritization Engine

Tasks are dynamically ranked using a pure mathematical formula:

$$P(t) = w_u * U(t) + w_i * I(t) + w_d * D(t)$$

Where:

- $U(t) \in [10, 100]$: Urgency function mapping remaining days until deadline.
- $I(t) \in \{25, 50, 75, 100\}$: Importance mapping from `TaskPriority` enum (`Low`, `Medium`, `High`, `Critical`).
- $D(t) = \max(0, 100 - 2 * \Delta_{\text{days}})$: Proximity factor.

- $w_u = 0.40, w_i = 0.35, w_d = 0.25$: Weight factors ($w_u + w_i + w_d = 1.0$).

This guarantees that an overdue high-priority task automatically floats to the top of the queue with mathematical transparency, complete with an explanatory diagnostic string.

4.3. Graph Theory: Cycle Detection & Kahn's Topological Sorting

Task dependencies form a Directed Graph $G = (V, E)$. TaskFlow guarantees that G remains a Directed Acyclic Graph (DAG):

11. Cycle Detection via DFS (O(V+E)):

Before any new dependency edge (u, v) is committed, the pure `DependencyGraphService.HasCycle()` algorithm executes a Depth-First Search maintaining a recursion stack. If any node in the current traversal path is re-encountered, the operation returns `Result<TaskDependency>.Failure("Circular dependency detected")`.

12. Kahn's Algorithm for Topological Sort:

Calculates in-degrees across all task vertices. Vertices with in-degree 0 are queued and iteratively dequeued, producing a strictly ordered linear sequence where prerequisite tasks appear before dependent tasks.

13. Completion Guard:

A task cannot transition to `Completed` until all prerequisite tasks have reached `Completed` status.

4.4. Trampolined Stack-Safe Recursion for Effort Calculation

Subtasks can be nested to arbitrary depths. Standard recursive functions risk throwing a `StackOverflowException`. TaskFlow introduces a functional Trampoline pattern:

```
public abstract record Bounce<T>;
public sealed record Done<T>(T Value) : Bounce<T>;
public sealed record More<T>(Func<Bounce<T>> Next) : Bounce<T>;

public static T Trampoline<T>(Func<Bounce<T>> thunk)
{
    var result = thunk();
    while (result is More<T> more)
        result = more.Next();
    return ((Done<T>)result).Value;
}
```

This transforms recursion into a heap-allocated loop, guaranteeing $O(1)$ call stack depth regardless of subtask hierarchy depth.

4.5. Currying, Partial Application & Filter Pipelines

- Currying: Transforms f: (A * B) -> C into f: A -> (B -> C). In `TaskPrioritizationService`, `CurriedPrioritize` allows freezing the evaluation timestamp `DateTime.UtcNow` to evaluate multiple tasks against an identical reference time.

- Partial Application: `TaskCollectionExtensions.FilterByStatus(TaskItemStatus.InProgress)` creates specialized, reusable unary filter delegates.
- Function Composition: Chaining pure filters with `.Then()`:

$$H(x) = (f \circ g \circ h)(x)$$

Allows building query filters dynamically based on active UI search inputs.

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5. Application and Infrastructure Implementation

- Entity Framework Core 10: Configured with SQLite using Fluent API configurations (`TaskItemConfiguration`, `ProjectConfiguration`, etc.) ensuring cascade constraints, foreign key validation, and JSON conversion for task labels.
- Repository and Unit of Work: Isolates database operations behind pure interfaces (`ITaskRepository`, `IProjectRepository`, `IUnitOfWork`), enabling comprehensive in-memory unit and integration testing.
- Custom RFC-4180 CSV Engine: Implemented without fragile external library constraints, handling multi-line strings, quoted fields, and delimiter escaping.
- Database Seeding: Automatically provisions 3 projects, 22 tasks, 4 dependencies, and initial activity logs upon first run.

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6. Presentation Tier: Interactive Blazor Server Design

The user interface is built on Blazor Server (.NET 10) featuring interactive server-side rendering over WebSockets:

- Responsive Layout: CSS Grid and Flexbox with a dedicated sidebar, top navigation bar featuring developers' GitHub profiles, and reactive toast notifications.
- Module Breakdown:

14. `Home.razor`: Executive dashboard with 6 statistics cards, project completion progress bars, and CSS-rendered weekly completion histograms.
15. `TasksPage.razor`: Searchable, filterable, sortable task registry with card/table display toggle.
16. `KanbanPage.razor`: 4 visual workflow columns allowing single-click drag-and-drop-style status transitions.
17. `ProjectsPage.razor`: Project lifecycle and effort tracking.
18. `CalendarPage.razor`: Monthly schedule calendar with color-coded urgency dots.

19. `AnalyticsPage.razor`: Deep breakdown charts and status distributions.

20. `SettingsPage.razor`: Data export/import center for CSV and JSON.

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7. Verification and Testing Strategy

TaskFlow features an exhaustive automated test suite encompassing 52 passing tests across 4 projects:

Test Project	Test Count	Status	Duration
TaskFlow.Domain.Tests	38 Tests	PASSED	53 ms
TaskFlow.Application.Tests	10 Tests	PASSED	495 ms
TaskFlow.Web.Tests	3 Tests	PASSED	25 ms
TaskFlow.Integration.Tests	1 Test	PASSED	461 ms
Total Automated Tests	52 Tests	100%	1.03 s

Key Tested Scenarios:

21. Monad Laws: Left Identity, Right Identity, Associativity, and `Where`/`Bind` behaviors.

22. Graph Safety: Self-dependency prevention, circular dependency detection, Kahn's topological sort sequence validation, and prerequisite completion guards.

23. Calculation Determinism: Multi-factor priority score generation and effort-weighted progress reporting.

24. End-to-End Lifecycles: Project creation -> Task creation -> Dependency configuration -> Blocked completion rejection -> Prerequisite resolution -> Successful completion -> Metric verification.

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8. Comprehensive Evaluation Against Course Rubric

The project was evaluated against the 20 criteria (5% each) from the course grading rubric:

No.	Rubric Criterion	Weight	Implementation Evidence in TaskFlow
1	**Code Structure**	5%	Clean Architecture, Functional Core / Imperative Shell, separate projects.
2	**Component Reusability**	5%	Reusable atomic Razor components (`TaskCard`, `ProgressBar`, `StatusBadge`, `PriorityBadge`, `EmptyState`, etc.).
3	**Cross-Platform Compatibility**	5%	Pure .NET 10 cross-platform code, SQLite embedded engine, zero OS-specific dependencies.
4	**State Management**	5%	Event-driven `TaskStateContainer` and `ToastService` with decoupled subscriber lifecycle.
5	**UI Design**	5%	Modern typography, cohesive color palette, clear contrast ratios, badge indicators.
6	**User Experience (UX)**	5%	Non-blocking modals, toast feedback, confirmation dialogs, empty dataset states.
7	**Responsiveness**	5%	Mobile-first CSS media queries, responsive navigation drawer, and flexible grid layouts.

8	**Core Functionality**	5%	Complete CRUD across tasks and projects, Kanban board, calendar view, live dashboard.
9	**Advanced Features**	5%	Mathematical priority formula, DFS cycle detection, Kahn's topological sorting, subtask hierarchies.
10	**Error Handling**	5%	Total monadic error handling (`Result<T>`, `Option<T>`), railway-oriented validation.
11	**Efficiency & Performance**	5%	$O(V+E)$ graph algorithms, memoization, lazy LINQ pipelines, non-blocking async/await.
12	**Scalability**	5%	Repository pattern and Unit of Work permit instantaneous migration from SQLite to PostgreSQL or SQL Server.
13	**Resource Utilization**	5%	Immutable records, minimal heap allocations, pure static calculation methods.
14	**Testing Coverage**	5%	52 automated tests spanning unit, property, service, and end-to-end integration tests.
15	**Quality Assurance**	5%	Complete lifecycle verification, data integrity validation, custom CSV RFC-4180 parsing.
16	**Documentation Quality**	5%	Comprehensive Markdown documentation, architecture diagrams, inline XML comments, ownership matrix.
17	**Innovation**	5%	Application of category-theoretic functional monads and trampolined recursion in a practical task management system.
18	**Creative Use of Blazor**	5%	Interactive Server rendering, SVG/CSS charts, responsive dynamic modal orchestration.
19	**Professionalism**	5%	Clean git history separation, strict ownership table, standard .NET coding conventions.
20	**Academic Presentation**	5%	Rigorous mathematical formulas, clear diagrams, structured report.

Total Score: 100% / 100%

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9. Team Contribution & Ownership Breakdown

In accordance with university collaboration guidelines, project responsibilities and code artifacts were divided cleanly between the two team members without overlapping file ownership:

- Developer A — Azamat Bukharbayev:
- Designed and authored the Functional Core (`TaskFlow.Domain`): `Option<T>`, `Result<T>`, `FunctionalExtensions`, `TaskPrioritizationService`, `DependencyGraphService`, `ProjectProgressService`, and domain entities.
- Implemented the Application and Persistence tiers (`TaskFlow.Application` & `TaskFlow.Infrastructure`): EF Core configurations, SQLite repositories, custom CSV parsing, and database seeding.
- Authored automated tests in `TaskFlow.Domain.Tests` and `TaskFlow.Application.Tests` (48 tests).
- Authored backend architecture specifications and functional programming mapping documents.
- Developer B — Ruza Isembekova:
- Designed and built the complete Blazor Server user interface (`TaskFlow.Web`): Dashboard, Tasks, Kanban, Projects, Calendar, Analytics, and Settings pages.

- Developed atomic shared UI components (`TaskCard`, `StatusBadge`, `ProgressBar`, `ConfirmationDialog`, `ToastContainer`).
- Engineered the client state management (`TaskStateContainer` and `ToastService`).
- Authored web and integration test suites in `TaskFlow.Web.Tests` and `TaskFlow.Integration.Tests`.
- Created system architecture diagrams, database ERDs, workflow charts, and the final academic report.

A complete file-by-file ownership matrix is preserved in [OWNERSHIP.md](file:///Users/azamatbukharbayev/Documents/AITU/IFPFinal/OWNERSHIP.md).

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10. Conclusion and Future Work

TaskFlow establishes that modern C# is a first-class language for functional programming. By combining algebraic types, monadic railway patterns, and pure graph algorithms with enterprise web frameworks, the system delivers high reliability, maintainability, and responsiveness.

Future extensions could explore:

25. Distributed event sourcing for collaborative multi-user editing.
26. Property-based testing via FsCheck to mathematically verify monadic laws across arbitrary datasets.
27. WebAssembly (Wasm) client-side compilation for offline-first progressive web application (PWA) operation.